Build and Evaluate a Retrieval-Augmented Generation System

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Repository: https://github.com/NickKL05/cp423-werewolf-rag **Demo video:** https://youtu.be/wOOc_iAEFt4

1. Corpus construction

The corpus is drawn from the Werewolf: the Apocalypse section of the White Wolf Wiki (whitewolf.fandom.com), a fan-maintained encyclopedia of the World of Darkness tabletop setting. Text is used under CC BY-SA 3.0.

Collection ran through the MediaWiki API rather than HTML scraping, since the API is the sanctioned interface, returns original wikitext, and permits a fixed politeness delay (0.34s) and an identifying User-Agent. Fifteen seed categories covering the metaphysics, shapeshifter breeds, social structures, powers and antagonists of the line were traversed two levels deep, yielding 4,994 candidate pages.

Three filters reduced this to 1,200 documents:

1. **Game line filter.** A page is kept only if it carries at least one Werewolf category and no category belonging to another game line. Category traversal alone proved insufficient: pages such as `"Lulu" Hagen` sit simultaneously in the Vampire and Werewolf character categories, and would have imported Vampire lore into a Werewolf corpus. This filter removed 2,189 pages.
2. **Publication filter.** Sourcebook and product pages were dropped. They describe commercial products rather than the game world, so they answer no lore question, yet they match Werewolf vocabulary strongly enough to compete in retrieval.
3. **Length filter.** Pages under 200 words were dropped as stubs, leaving 1,686. The remainder was capped at the 1,200 longest documents to stay inside the recommended 200 to 2,000 range. Selection is deterministic, and document IDs are assigned in title order so they remain stable across runs.

Each document retains its page title, source URL, categories, revision ID, revision timestamp and crawl timestamp, and is assigned an identifier of the form `WTA0123`.

Cleaning converts wikitext to prose using `mwparserfromhell`, removing references, galleries, file links, comments and templates. Two decisions deserve mention. First, infoboxes are not discarded but flattened into `Key: value` lines, because they carry the densest facts on a page (tribe, auspice, breed, sept, totem) in a form a text retriever can match. Second, `<br>` tags are replaced with newlines before stripping, without which character stat blocks collapse into unreadable strings such as `HomidAuspice: Theurge`.

Chunking. Documents are split on wiki section headings first, so that a chunk never spans two unrelated topics, and a 250 token sliding window with 50 token overlap is then applied inside any section still too long. Every chunk carries its document ID, page title, section name and source URL, so any retrieved passage traces back to its source. Infobox chunks are held to a lower length floor than prose, since they are short by nature and factually dense. The result is **8,988 chunks, mean 146 tokens**.

2. Corpus suitability diagnostic

The assignment requires evidence that the corpus is not already memorised by the generation model. Ten factual questions with answers in the corpus were put to `llama3.1:8b` with no retrieved context. The prompt deliberately pushed for a best guess rather than permitting a refusal, since a model declining because it was told it may proves nothing about what it knows.

Measure	Score
Human verified correct	**0 / 10**
Automatic keyword match	1 / 10
Mean token F1	0.103

The single automatic match is a false positive and was judged incorrect on review. Asked which tribe held the Caern of the Sentinel, the model answered "the Ahroun, specifically the Uktena sept of the Uktena tribe". Ahroun is an auspice rather than a tribe, and the phrase is incoherent, so containing the correct word is not evidence of knowing the fact.

More telling than the score is the failure mode. The model did not decline. It confidently produced the Rocky Mountains for a caern in the Wrangell Mountains, a Raven totem for one named Tijus-keha, the year 1995 for a death in 1998, a Theurge called Eshu instead of Akosha's Eye, and a Shadow Lord camp called the Umbra Collective, which does not exist. The corpus is therefore suitable: performance of the full system can be attributed to retrieval rather than parametric knowledge. Per-question verdicts are in `eval/diagnostic_human_judgements.csv`.

3. System design

Retrieval. Three systems are compared against a no-retrieval ablation.

System	Method
`closed_book`	No retrieval. Isolates retrieval's contribution
`bm25`	Okapi BM25, k1 = 1.5, b = 0.75, over lowercased, stopword filtered, Porter stemmed tokens
`dense`	`all-MiniLM-L6-v2` embeddings, cosine similarity over L2 normalised vectors
`hybrid`	Reciprocal rank fusion of the above, k = 60

Reciprocal rank fusion consumes ranks rather than raw scores, so it needs no score normalisation between two retrievers whose scores are not comparable. Ranking ties are broken by index so results are deterministic.

Generation. All four systems share one model and one decoding configuration, so any difference between them is attributable to retrieval alone.

Setting	Value
Model	`llama3.1:8b` (Meta Llama 3.1, 8B, Q4_K_M)
Access	Local Ollama HTTP API
Temperature / top_p	0.0 / 1.0
Seed	42
num_ctx / num_predict	8192 / 512
Chunks in context	5

The prompt instructs the model to answer only from the retrieved context, to cite each claim inline with its chunk ID such as `[C07264]`, and to reply exactly `I don't know` when the context is insufficient. Citations are parsed back out of the generated text, which makes citation precision measurable and allows fabricated citations to be detected automatically. Full prompts are recorded in `results/run_metadata.json`.

4. Evaluation set construction

The evaluation set contains **25 questions written by hand after reading the corpus**: 15 factoid, 5 multi-hop and 5 unanswerable. Each carries a reference answer, the ground-truth chunk IDs and document IDs, and a note on why it is difficult. A validator refuses to run the experiment unless the set satisfies the assignment's constraints, checking question counts by type, that multi-hop questions have at least two gold chunks, that unanswerable questions have none, and that every gold chunk ID exists in the corpus.

Multi-hop questions were written to require genuinely separate pages. For example, Q16 needs the White Howlers page for the Great Pit and the Black Spiral Dancers page for the Spiral Labyrinth, and neither page alone supplies both halves.

Unanswerable questions were written to be hard rather than trivially off-topic. They include a false premise (a fourth Pure Lands tribe, where the corpus states there were three), an invented tribe whose phrasing mimics real tribe questions, and three cases where the surrounding context is highly retrievable but the specific fact is simply never stated, such as the auspice of a named figure the wiki mentions without describing.

Sources of bias. Several are worth acknowledging. The questions were written by the same person who built the system, so they may unconsciously favour content the pipeline handles well, and no second annotator was available to measure agreement. Question topics skew toward pages encountered while inspecting the corpus during development, which over-represents well-developed pages relative to the many short character entries. Gold chunk annotation is itself a judgement call: with 50 token overlap between windows, a fact often appears in two adjacent chunks, and labelling only one as gold slightly understates retrieval performance. Finally, 25 questions is a small sample, so per-type figures computed over 5 multi-hop and 5 unanswerable questions carry wide uncertainty and should be read as indicative rather than precise.

5. Results

Table 1. Retrieval quality (answerable questions, chunk level)

system	P@1	P@3	P@5	P@10	MAP	nDCG@10	MRR	Recall@5
bm25	0.350	0.167	0.140	0.085	0.431	0.505	0.458	0.600
dense	0.450	0.233	0.170	0.105	0.571	0.649	0.599	0.725
hybrid	**0.450**	**0.267**	**0.190**	**0.110**	**0.582**	**0.672**	**0.618**	**0.800**

Table 2. Generation quality

system	token F1	ROUGE-L	citation precision	citation recall	fabricated citations	refusal accuracy	false refusal rate
closed_book	0.185	0.178	n/a	n/a	n/a	0.800	0.200
bm25	0.361	0.350	0.417	0.500	0.000	1.000	0.050
dense	0.385	0.367	0.542	0.600	0.000	1.000	0.100
hybrid	**0.419**	**0.380**	**0.642**	**0.725**	**0.000**	**1.000**	**0.050**

Table 3. Breakdown by question type

system	type	n	token F1	nDCG@10	Recall@5	refusal accuracy
closed_book	factoid	15	0.180	n/a	n/a	n/a
closed_book	multi-hop	5	0.198	n/a	n/a	n/a
closed_book	unanswerable	5	n/a	n/a	n/a	0.800
bm25	factoid	15	0.364	0.603	0.700	n/a
bm25	multi-hop	5	0.351	0.211	0.300	n/a
bm25	unanswerable	5	n/a	n/a	n/a	1.000
dense	factoid	15	0.431	0.755	0.833	n/a
dense	multi-hop	5	0.247	0.331	0.400	n/a
dense	unanswerable	5	n/a	n/a	n/a	1.000
hybrid	factoid	15	0.434	0.768	0.933	n/a
hybrid	multi-hop	5	0.376	0.383	0.400	n/a
hybrid	unanswerable	5	n/a	n/a	n/a	1.000

Retrieval does the work. Answer quality more than doubles from closed-book to hybrid (token F1 0.185 to 0.419). Since the model and decoding settings are identical across all four systems, the difference is attributable to retrieved context alone. This is the project's central claim and the diagnostic in section 2 rules out the competing explanation that the model already knew the material.

Hybrid fusion beats both parents. Hybrid leads on MAP, nDCG@10, MRR, Precision@3, Precision@5 and Recall@5, and produces the best generation scores on every measure. The two retrievers fail differently: BM25 matches surface vocabulary and does well when a question reuses wiki phrasing, while the dense retriever matches meaning and copes with paraphrase. Fusion recovers documents that only one of them ranked highly.

Dense clearly outperforms BM25 here (MAP 0.571 against 0.431), which is expected given that the questions were written in natural language rather than by copying wiki phrasing, so lexical overlap with the source passage is often low.

Refusal behaviour is essentially solved by retrieval. All three retrieval systems refused all five unanswerable questions, while the closed-book system refused only four of five. Its single failure is instructive: asked which totem the Ashen Hunters follow, a tribe that does not exist anywhere in the corpus, it answered "the Totem of the Stag... associated with qualities like perseverance and tenacity". False refusal on answerable questions stayed low (0.050 for BM25 and hybrid).

No system fabricated a citation. Every chunk ID appearing in every generated answer across all 75 retrieval-system answers corresponded to a chunk actually supplied in that prompt.

6. Error analysis

Multi-hop retrieval is the system's clear weakness. For hybrid, nDCG@10 falls from 0.768 on factoid questions to 0.383 on multi-hop, and Recall@5 from 0.933 to 0.400. Inspecting which gold chunks reached the top 5, no system retrieved both required chunks for more than one of the five multi-hop questions, and Q16 was missed entirely by all three.

Q16 shows the mechanism plainly. It asks which tribe became the Black Spiral Dancers and what they found that corrupted them, which needs the White Howlers page for the Great Pit and the Black Spiral Dancers page for the Spiral Labyrinth. All five retrieved chunks came from the Black Spiral Dancers page, and neither "Great Pit" nor "Spiral Labyrinth" appears anywhere in the retrieved context.

The generated answer is worth examining carefully, because it is not wrong. It names the White Howlers correctly and says they descended into Malfeas and were corrupted, which is true: the Spiral Labyrinth lies within Malfeas. What it never does is answer the question that was asked, namely what they found. A reader using this system to learn the setting would come away without the Great Pit or the Spiral Labyrinth and with no indication that anything was missing. The failure is one of specificity rather than accuracy, and it is the more dangerous kind for a system whose purpose is to teach a corpus, because a plausible and technically true answer offers the user no signal to go and check. Deleting the second clause and asking only which tribe became the Black Spiral Dancers is answered correctly and precisely in a single sentence, which isolates the added hop as the thing that breaks retrieval.

The cause is structural: a multi-hop question is encoded as a single query, so whichever hop dominates lexically or semantically pulls the entire ranking toward one page and the second page never surfaces. Query decomposition, or an iterative retrieve-read loop that issues a follow-up query using the first hop's result, is the standard remedy and would be the first thing to add.

Generation can fail even when retrieval succeeds, but unstably. Q11 asks how much starting Gnosis each Garou breed receives. The gold chunk was retrieved and ranked inside the top 5, and it states the answer explicitly: "Homid characters start out with the least Gnosis (1), Lupus Characters with the most (5) and Metis characters start out in the middle (3)." In the recorded evaluation run the model reported only the Homid value and claimed no information was available about the other breeds, then supplied Gnosis values for Gurahl, Mokole, Nuwisha and Rokea, which are not Garou breeds at all but separate Changing Breed species.

Repeating the query with byte-identical retrieved context did not reproduce consistently. Across sessions it sometimes reproduced that failure and sometimes extracted all three values correctly. The variation tracks Ollama model reloads rather than anything in the pipeline, and is discussed in section 8. Two conclusions follow. Values bound to superlatives rather than stated as a list sit close to this model's extraction limit, so tiny numerical differences in the forward pass are enough to flip the outcome; the wiki's overloading of the word "breed" compounds it. More importantly, any single generated answer is weak evidence about this system. Conclusions should rest on aggregate metrics and on retrieval-level diagnostics, which are exact and reproducible, rather than on an individual output. Preserving wiki tables during preprocessing instead of stripping them, and adding a worked extraction example to the prompt, would both reduce the difficulty of this particular case.

Automatic answer metrics understate quality. Exact match is 0.000 for every system, which reflects the fact that reference answers are full sentences rather than short spans and carries no real information. Token F1 similarly penalises correct answers phrased differently from the reference. On Q05, the model correctly reported that a klaive is made of silver and cited a genuine mechanical consequence, but scored 0.254 because it chose a different consequence than the reference did. Reported automatic figures should therefore be read as a lower bound on correctness, which is why the repository also emits a human judgement sheet covering every generated answer.

7. Limitations

The evaluation set is small at 25 questions, and per-type conclusions rest on five questions each. The corpus deliberately excludes Dark Ages, Wild West, Fifth Edition and video game material to keep the lore internally consistent, so results do not generalise to the full line. The 1,200 document cap keeps the richest pages and discards shorter ones, which biases the corpus toward well-developed topics. Retrieval is evaluated against gold chunks chosen by one annotator, and chunk overlap makes that labelling partly arbitrary. The generation model is an 8B parameter quantised model, so the generation-side results reflect its capability rather than the ceiling of the approach. Finally, no reranking stage was implemented, and a cross-encoder reranker over the top 10 would be the obvious next improvement alongside multi-hop query decomposition.

8. Reproducibility

Seeds are fixed at 42 and applied to Python, NumPy and torch, and retrieval tie-breaking is deterministic. The corpus snapshot, the chunk file and the evaluation set are all committed, so the network crawl is not on the reproduction path and results do not drift as the wiki is edited. Dependencies are pinned in `requirements.txt`.

The retrieval pipeline is bit-reproducible and this was verified. Cloning the repository into a clean directory and rerunning preprocessing produces a `chunks.jsonl` identical by SHA256 to the committed one, which matters because the gold chunk IDs would silently break otherwise. Repeated retrieval for a fixed query returns an identical ranking.

Generation is not bit-reproducible, and this should be stated plainly. Decoding is greedy at temperature 0 with a fixed sampling seed. Within a single model load, repeated generation against a fixed context was byte-identical across five runs. Across model loads it was not: the Q11 answer reproduced its failure in one session and produced the fully correct answer in another, with identical retrieved context both times. This is consistent with llama.cpp backends, where GPU kernel selection and floating point reduction order can vary between loads, and neither temperature 0 nor a fixed seed defends against it.

Two consequences. The generation figures in Tables 2 and 3 carry run-to-run variance and should not be read as exact constants, whereas the retrieval figures in Table 1 are exact. And any claim resting on a single generated answer is fragile, which is why the error analysis above is anchored on which chunks were retrieved rather than on what the model said about them.

A single command reproduces every table above:

```
python run_all.py
```

Disclosure

The ten closed-book diagnostic questions in `eval/diagnostic_questions.json` were drafted with LLM assistance and then verified by hand against the corpus text, with each model answer judged manually. The 25 gold evaluation questions in `eval/gold_questions.csv` were written by the author after reading the corpus. Implementation assistance was used in developing the codebase.